CSEP 590tv: Quantum Computing

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July 13, 2005





Today's Menu

Administrivia

Partial Measurements

Circuit Elements

Deutsch's Algorithm

Quantum Teleportation

Superdense Coding

Administrivia

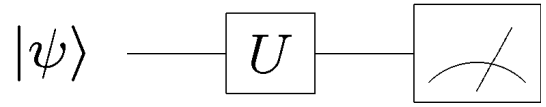
Hand in HW #2

Pick up HW #3 (due July 20)

HW #1 solution available on website

Recap

Unitary rotations and measurements in different basis



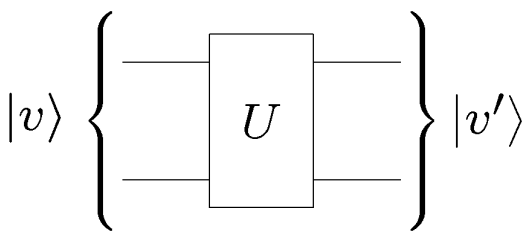
Two qubits.

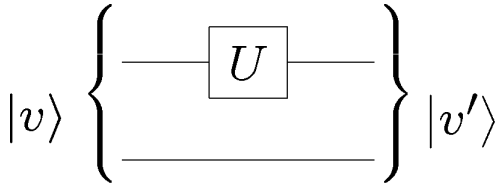
$$|v\rangle = v_{00}|00\rangle + v_{01}|01\rangle + v_{10}|10\rangle + v_{11}|11\rangle$$

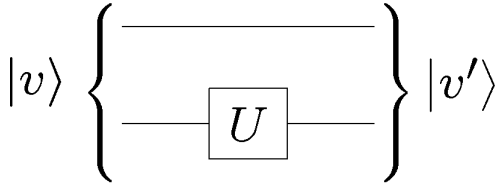
Separable versus Entangled.

$$|v\rangle = |a\rangle \otimes |b\rangle \qquad |v\rangle = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

Single qubit versus two qubit unitaries



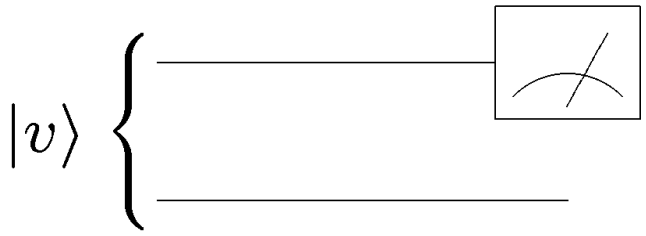




Partial Measurements

Say we measure one of the two qubits of a two qubit system:

$$|v\rangle = v_{00}|00\rangle + v_{01}|01\rangle + v_{10}|10\rangle + v_{11}|11\rangle$$



1. What are the probabilities of the different measurement outcomes?

2. What is the new wave function of the system after we perform such a measurement?

Matrices, Bras, and Kets

So far we have used bras and kets to describe row and column vectors. We can also use them to describe matrices:

Outer product of two vectors:

$$|v\rangle\langle w| = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \begin{bmatrix} w_1^* & w_2^* \end{bmatrix}$$

$$= \begin{bmatrix} v_1 w_1^* & v_1 w_2^* \\ v_2 w_1^* & v_2 w_2^* \end{bmatrix}$$

Example:

$$|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \qquad \langle w| = \begin{bmatrix} \frac{i}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$|v\rangle\langle w| = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{i}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}}\frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}}\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}}\frac{-1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{i}{2} & -\frac{1}{2} \\ \frac{i}{2} & -\frac{1}{2} \end{bmatrix}$$

Matrices, Bras, and Kets

We can expand a matrix about all of the computational basis outer products

$$M = \sum_{i,j=0}^{N-1} M_{i,j}|i\rangle\langle j| = \begin{bmatrix} M_{0,0} & \cdots & M_{N-1,0} \\ \vdots & \ddots & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 1 & i \\ -1 & -i \end{bmatrix}$$

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad |0\rangle\langle 1| = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$|1\rangle\langle 0| = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$M = |0\rangle\langle 0| + i|0\rangle\langle 1| - 1|1\rangle\langle 0| - i|1\rangle\langle 1|$$

Matrices, Bras, and Kets

We can expand a matrix about all of the computational basis outer products

$$M = \sum_{i,j=0}^{N-1} M_{i,j}|i\rangle\langle j| = \begin{bmatrix} M_{0,0} & \cdots & M_{N-1,0} \\ \vdots & \ddots & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

This makes it easy to operate on kets and bras:

$$M|v\rangle = \sum_{i,j=0}^{N-1} M_{i,j}|i\rangle\langle j|v\rangle$$

complex numbers

$$\langle w|M = \sum_{i,j=0}^{N-1} M_{i,j}\langle w|i\rangle\langle j|$$

Matrices, Bras, and Kets

Example:

$$M = \begin{bmatrix} 1 & i \\ -1 & -i \end{bmatrix}$$

$$M = |0\rangle\langle 0| + i|0\rangle\langle 1| - 1|1\rangle\langle 0| - i|1\rangle\langle 1|$$

$$|v\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

$$M|v\rangle = (|0\rangle\langle 0| + i|0\rangle\langle 1| - 1|1\rangle\langle 0| - i|1\rangle\langle 1|)\left(\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle\right)$$

$$= \frac{1}{2}|0\rangle + i\frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle - i\frac{\sqrt{3}}{2}|1\rangle$$

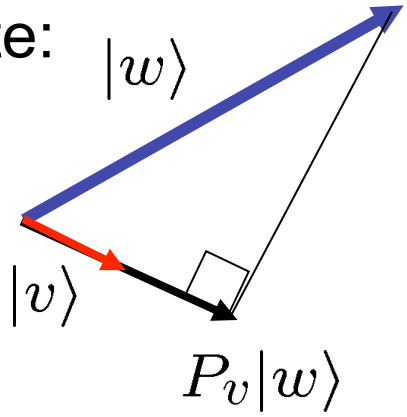
$$= \frac{1 + i\sqrt{3}}{2}|0\rangle - \frac{1 + i\sqrt{3}}{2}|1\rangle$$

Projectors

The projector onto a state $|v\rangle$ (which is of unit norm) is given by

$$P_v = |v\rangle\langle v|$$

Projects onto the state:



Note that $P_v|v\rangle = |v\rangle\langle v|v\rangle = |v\rangle$

and that

$$P_v|w\rangle = |v\rangle\langle v|w\rangle = (\langle v|w\rangle)|v\rangle$$

Example: $|v\rangle = |0\rangle \quad P_v = |0\rangle\langle 0|$

$$|w\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

$$P_v|w\rangle = |0\rangle\langle 0|\left(\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle\right) = \frac{1}{2}|0\rangle$$

Measurement Rule

If we measure a quantum system whose wave function is $|v\rangle$ in the basis $|w_i\rangle$, then the probability of getting the outcome $|w_i\rangle$ corresponding to is given by

$$Pr(|w_i\rangle) = |\langle w_i|v\rangle|^2 = \langle v|w_i\rangle\langle w_i|v\rangle = \langle v|P_{w_i}|v\rangle$$

where

$$P_{w_i} = |w_i\rangle\langle w_i|$$

The new wave function of the system after getting the measurement outcome corresponding to $|w_i\rangle$ is given by

$$|v'\rangle = \frac{P_{w_i}|v\rangle}{\sqrt{Pr(|w_i\rangle)}}$$

For measuring in a complete basis, this reduces to our normal prescription for quantum measurement, but…

Measuring One of Two Qubits

Suppose we measure the first of two qubits in the computational basis. Then we can form the two projectors:

$$P_0 \otimes I = |0\rangle\langle 0| \otimes I \qquad I = |0\rangle\langle 0| + |1\rangle\langle 1|$$

$$P_1 \otimes I = |1\rangle\langle 1| \otimes I$$

If the two qubit wave function is $|v\rangle$ then the probabilities of these two outcomes are

$$Pr(0) = \langle v|P_0 \otimes I|v\rangle$$

$$Pr(1) = \langle v|P_1 \otimes I|v\rangle$$

And the new state of the system is given by either

$$|v'\rangle = \frac{P_0 \otimes I|v\rangle}{\sqrt{Pr(0)}}$$

Outcome was 0

$$|v'\rangle = \frac{P_1 \otimes I|v\rangle}{\sqrt{Pr(1)}}$$

Outcome was 1

Measuring One of Two Qubits

Example $|v\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{\sqrt{2}}|11\rangle$

Measure the first qubit: $P_0 \otimes I = |0\rangle\langle 0| \otimes I \quad P_1 \otimes I = |1\rangle\langle 1| \otimes I$

$P_0 \otimes I = |0\rangle\langle 0| \otimes (|0\rangle\langle 0| + |1\rangle\langle 1|) = |00\rangle\langle 00| + |01\rangle\langle 01|$

$$Pr(0) = \langle v | P_0 \otimes I | v \rangle$$

$$P_0 \otimes I |v\rangle = (|00\rangle\langle 00| + |01\rangle\langle 01|)|v\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle$$

$$Pr(0) = \left(\frac{1}{2}\langle 00| + \frac{1}{2}\langle 01| + \frac{1}{\sqrt{2}}\langle 11|\right)\left(\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle\right)$$

$$Pr(0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \qquad |v'\rangle = \frac{P_0 \otimes I |v\rangle}{\sqrt{Pr(0)}} = \frac{\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle}{\frac{1}{\sqrt{2}}}$$

$$|v'\rangle = \frac{1}{\sqrt{2}}|00\rangle + \sqrt{\frac{1}{2}}|01\rangle = |0\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \sqrt{\frac{1}{2}}|1\rangle\right)$$

Instantaneous Communication?

Suppose two distant parties each have a qubit and their
joint quantum wave function is

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

If one party now measures its qubit,
then…

$$P_0 \otimes I = |0\rangle\langle 0| \otimes I \qquad Pr(0) = \frac{1}{2} \qquad |v'\rangle = |0\rangle \otimes |0\rangle$$

$$P_1 \otimes I = |1\rangle\langle 1| \otimes I \qquad Pr(1) = \frac{1}{2} \qquad |v'\rangle = |1\rangle \otimes |1\rangle$$

The other parties qubit is now either the $|0\rangle$ or $|1\rangle$

Instantaneous communication? NO.
Why NO? These two results happen with probabilities.

Correlation does not imply communication.

In Class Problem 1

You Are Now a Quantum Master

Important Single Qubit Unitaries

Pauli Matrices:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$ "bit flip"

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$ "phase flip"

$$\sigma_0 = I \qquad \sigma_1 = X \qquad \sigma_2 = Y \qquad \sigma_3 = Z \qquad \sigma_j^2 = I$$

$|0\rangle$ — X — $|1\rangle$

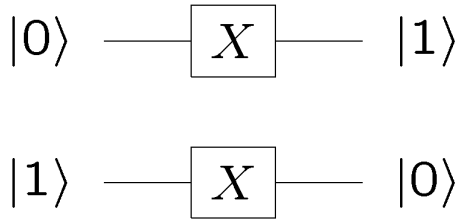
$|1\rangle$ — X — $|0\rangle$

"bit flip" is just the classical not gate

$$X\,Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = -iZ$$

$$Y\,X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = iZ$$

Important Single Qubit Unitaries



"bit flip" is just the classical not gate



Jacques Hadamard

Hadamard gate:

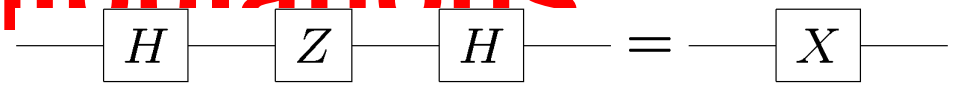
$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \qquad H^2 = I$$

$$HZH = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

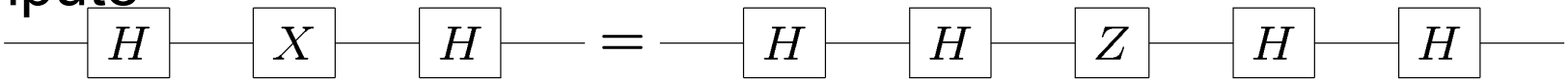
$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = X$$



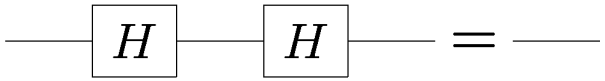
Single Qubit Manipulations



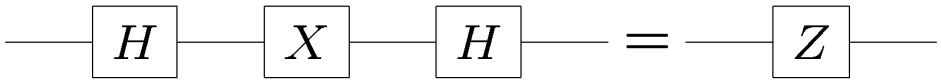
Use this to compute HXH



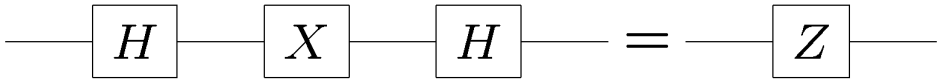
But $H^2 = I$

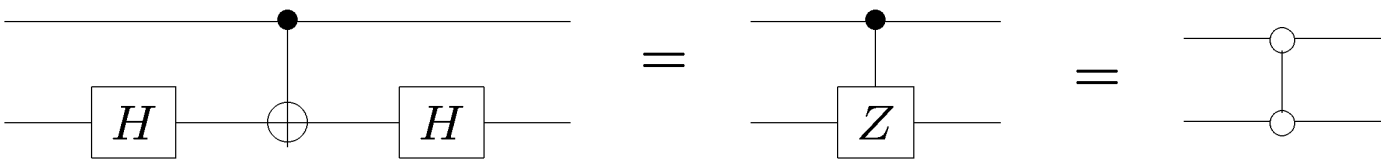


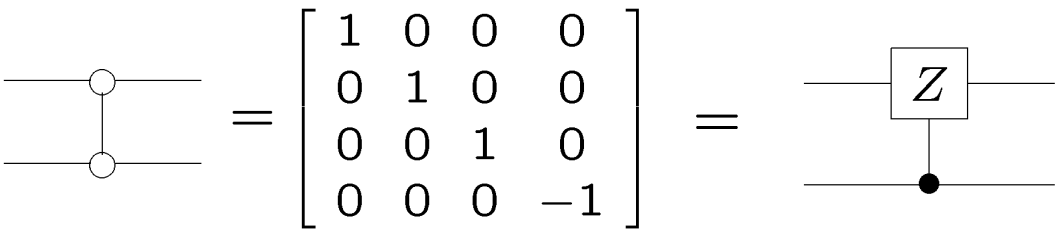
So that

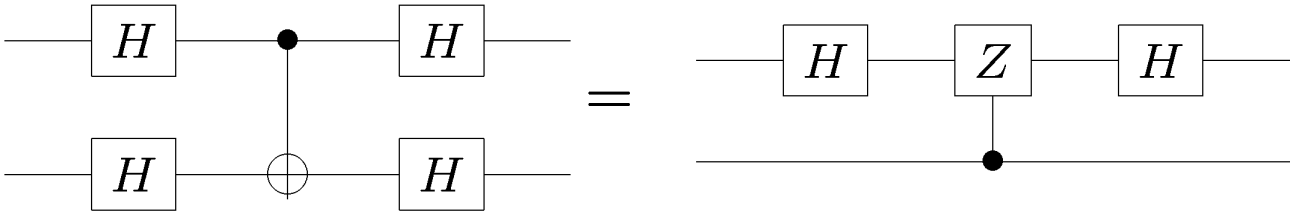


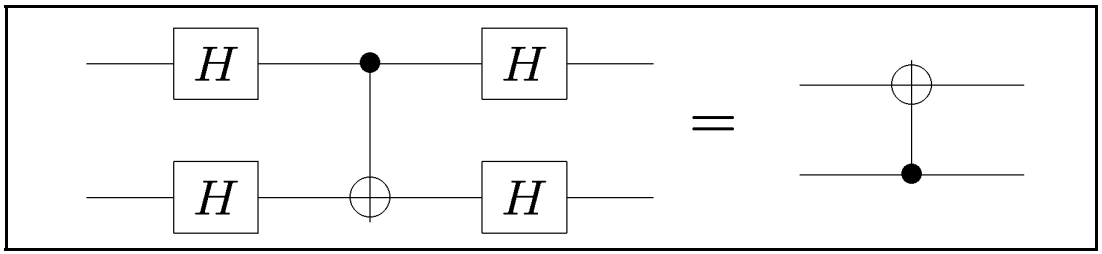
A Cool Circuit Identity

Using 



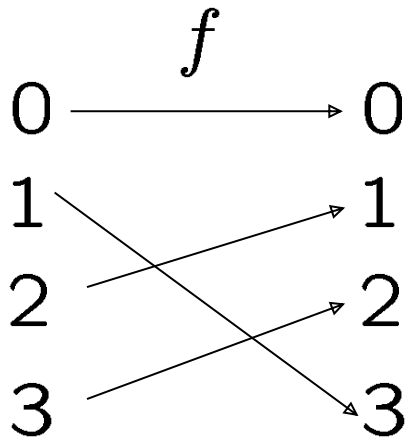






Reversible Classical Gates

A reversible classical gate on k bits is one to one function on the 2^k values of these bits.



Example :

$00 \rightarrow 00$
$01 \rightarrow 01$
$10 \rightarrow 11$
$11 \rightarrow 10$

reversibl

$00 \rightarrow 00$
$01 \rightarrow 00$
$10 \rightarrow 10$
$11 \rightarrow 11$

not

Reversible Classical Gates

A reversible classical gate on k bits is one to one function on the 2^k values of these bits.

We can represent reversible classical gates by a permutation matrix.

Permutation matrix is matrix in which every row and column contains at most one 1 and the rest of the elements are 0

Example: $00 \rightarrow 00$, $01 \rightarrow 01$, $10 \rightarrow 11$, $11 \rightarrow 10$

reversibl

$$\begin{array}{c} \\ 00 \\ 01 \\ 10 \\ 11 \end{array} \begin{array}{c} \begin{array}{cccc} 00 & 01 & 10 & 11 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

output

inpu t

Quantum Versions of Reversible Classical Gates

A reversible classical gate on k bits is one to one function on the 2^k values of these bits.

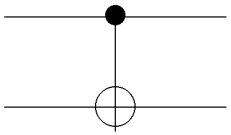
We can turn reversible classical gates into unitary quantum gates

Permutation matrix is matrix in which every row and column contains at most one 1 and the rest of the elements are 0

Use permutation matrix as unitary evolution matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

controlled-NOT



$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

David Speaks



David Deutsch 1985

"Complexity theory has been mainly concerned with constraints upon the computation of functions: which functions can be computed, how fast, and with use of how much memory. With quantum computers, as with classical stochastic computers, one must also ask 'and with what probability?' We have seen that the minimum computation time for certain tasks can be lower for Q than for T . Complexity theory for Q deserves further investigation."

Q = quantum computers
T = classical computers

Deutsch's Problem

Suppose you are given a black box which computes one of the following four reversible gates:

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
	constant		balanced

Deutsch's (Classical) Problem:
How many times do we have to use this black box to determine whether we are given the first two or the second two?

Classical Deutsch's Problem

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
	constant		balanced

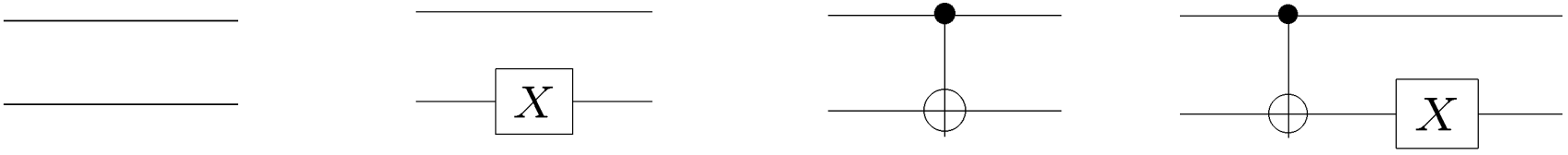
Notice that for every possible input, this does not separate the "constant" and "balanced" sets. This implies at least one use of the black box is needed.

Querying the black box with 00 and 10 distinguishes between
these two sets. Two uses of the black box are necessary and sufficient.

Classical to Quantum Deutsch

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit

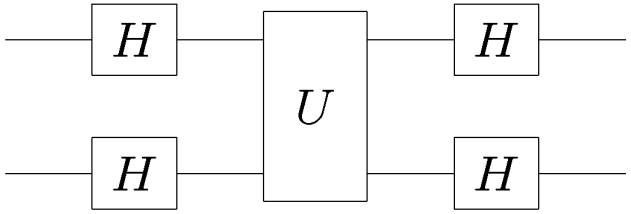
Convert to quantum gates

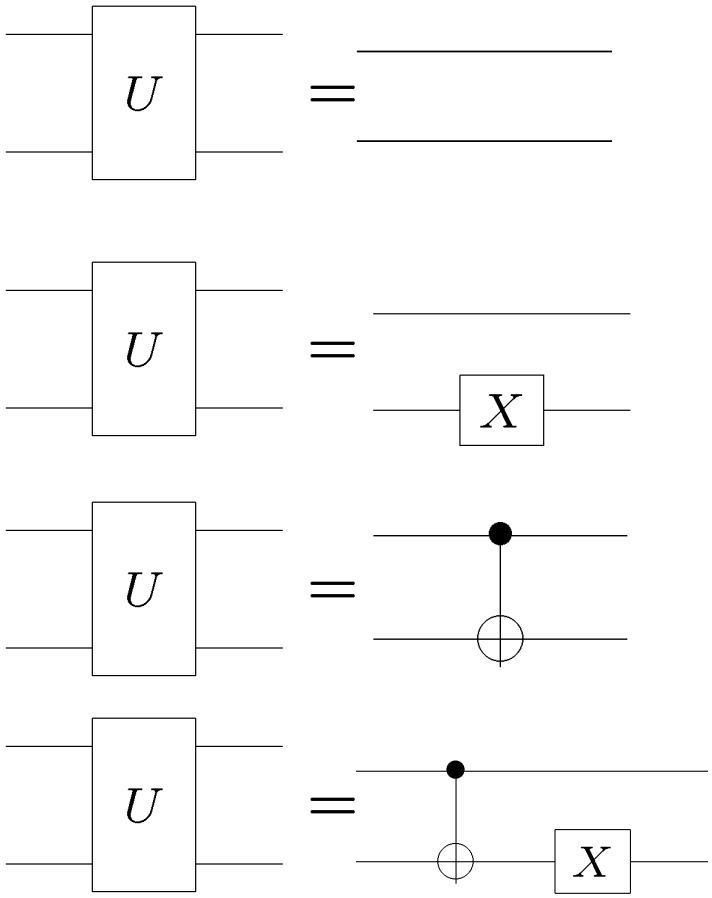


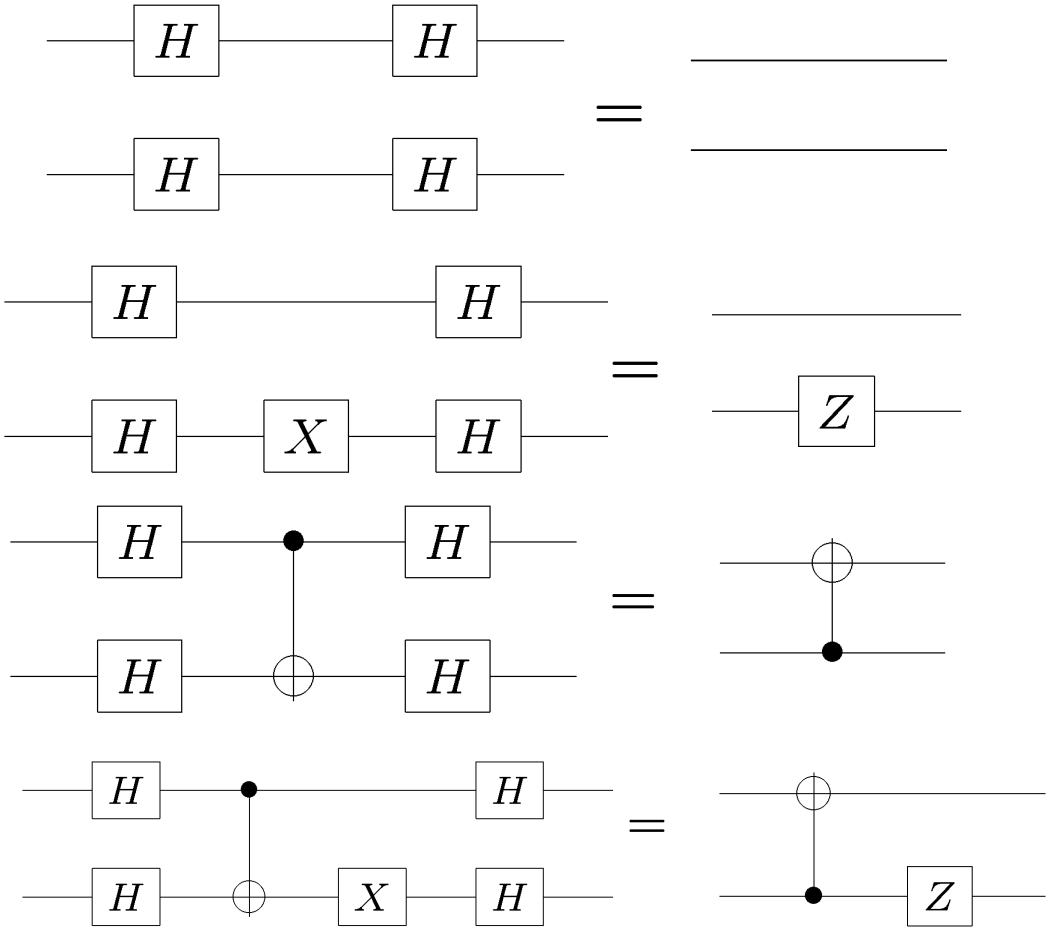
Deutsch's (Quantum) Problem:
How many times do we have to use these quantum gates to determine whether we are given the first two or the second two?

Quantum Deutsch

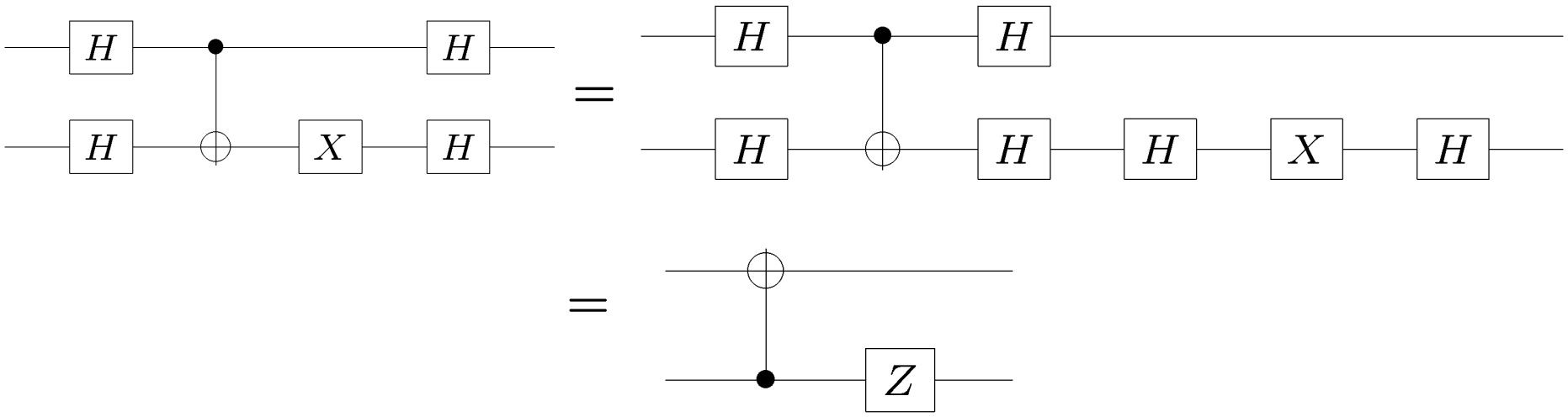
What if we perform Hadamards before and after the quantum gate:



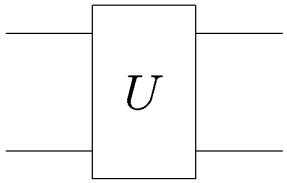


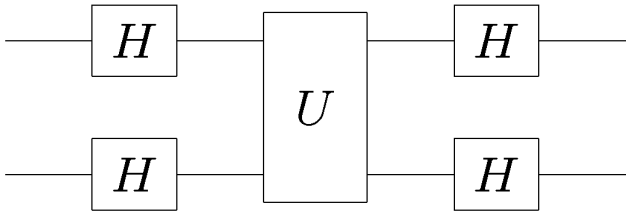


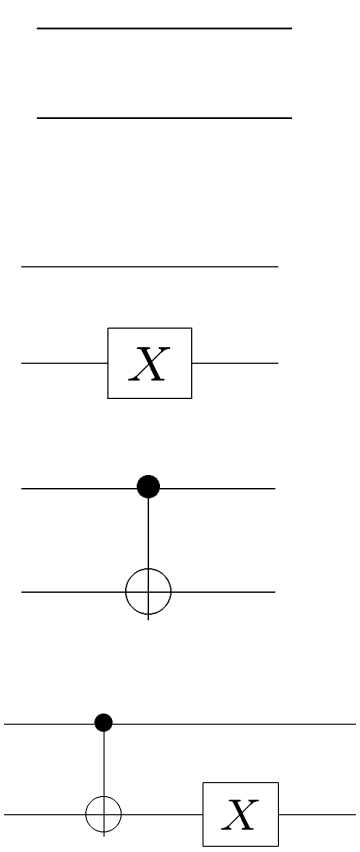
That Last One

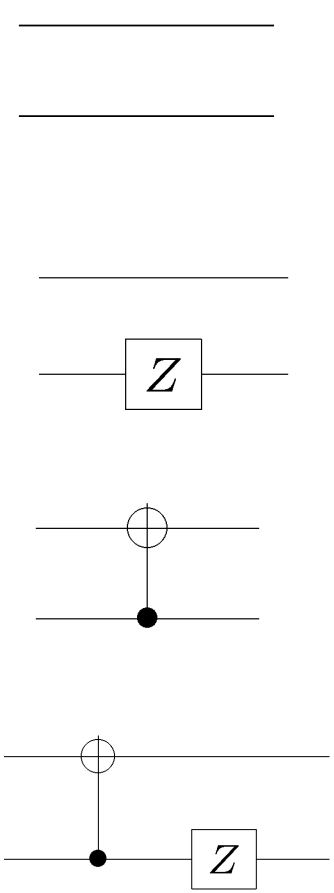


Agai
n

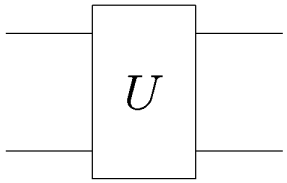


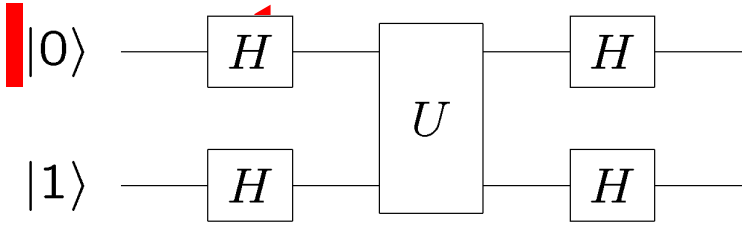


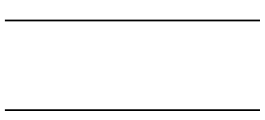


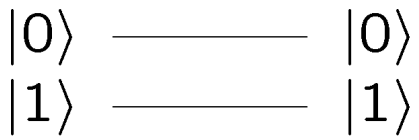


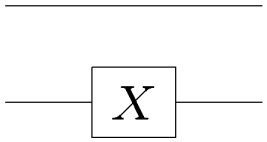
Some

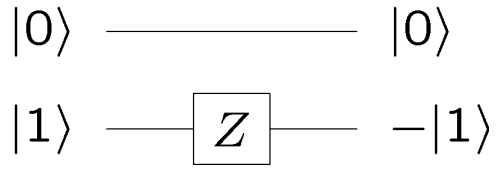


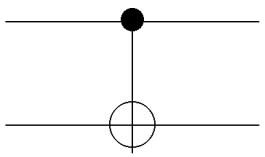


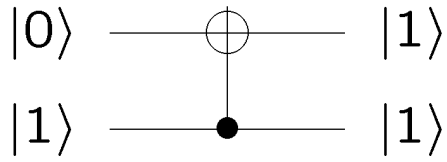


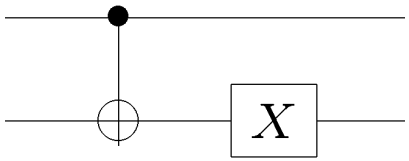


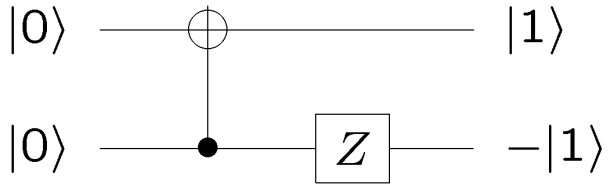




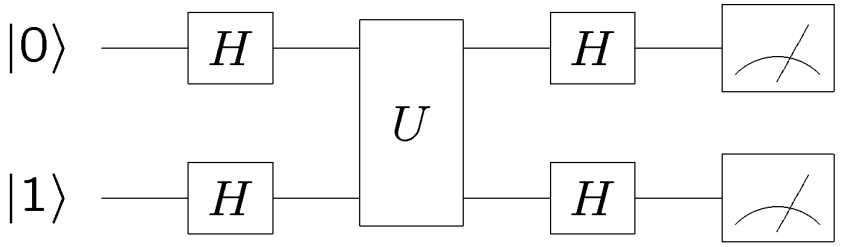


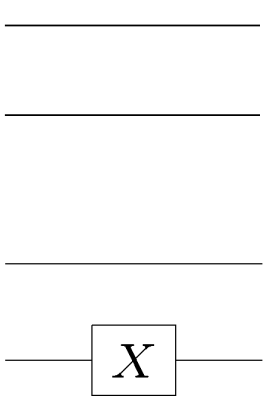


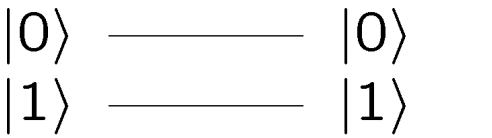


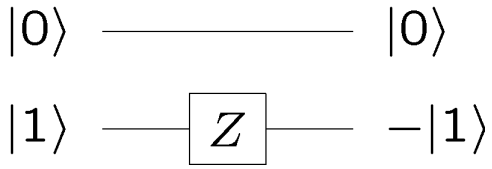


Quantum Deutsch

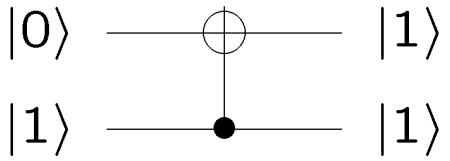


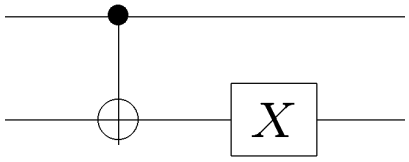


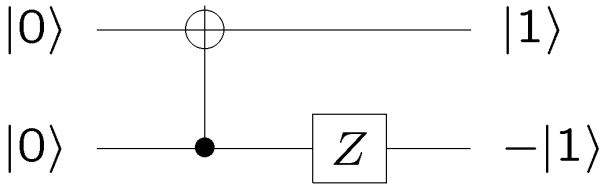




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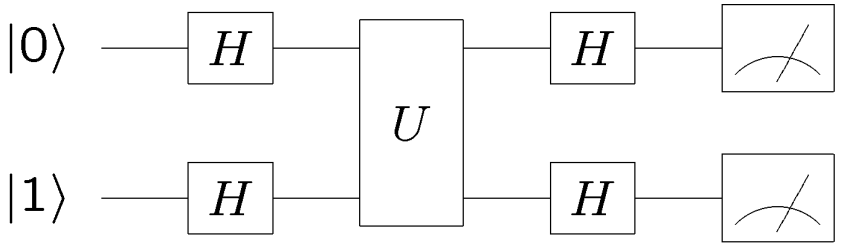






= 1

Quantum Deutsch



By querying with quantum states we are able to distinguish the first two (constant) from the second two (balanced) with
only one use of the quantum gate!

Two uses of the classical gates
Versus
One use of the quantum gate

first quantum speedup (Deutsch, 1985)

In Class Problem 2

Quantum Teleportation

Alice wants to send her qubit to Bob.
She does not know the wave function of her qubit.

Alic
e



Bob



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Can Alice send her qubit to Bob using classical bits?

Since she doesn't know $|\psi\rangle$ and measurements on her state do not reveal $|\psi\rangle$, this task appears impossible.

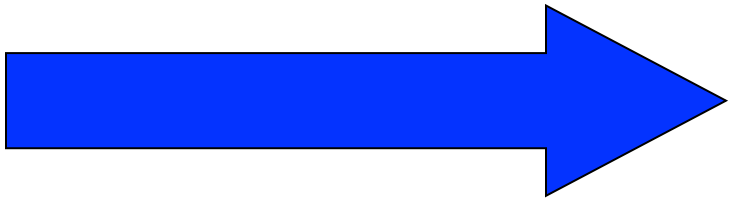
Quantum Teleportation

Alice wants to send her qubit to Bob.
She does not know the wave function of her qubit.

Alic
e



classical communication



Bob



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Suppose these bits contain information about $|\psi\rangle$

Then Bob would have information about $|\psi\rangle$ as well as the qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

This would be a procedure for extracting information from $|\psi\rangle$ without effecting the state

~~Quantum~~ Teleportation Classical

Alice wants to send her probabilistic bit to Bob using classical communication.

Alic
e



Bob



$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

She does not wish to reveal any information about this bit.

Classical Teleportation

(a.k.a. one time pad)

Alic
e



$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix}$$

$$e = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$

50 %
00
50 % 11

Bob



Alice and Bob have two perfectly correlated bits

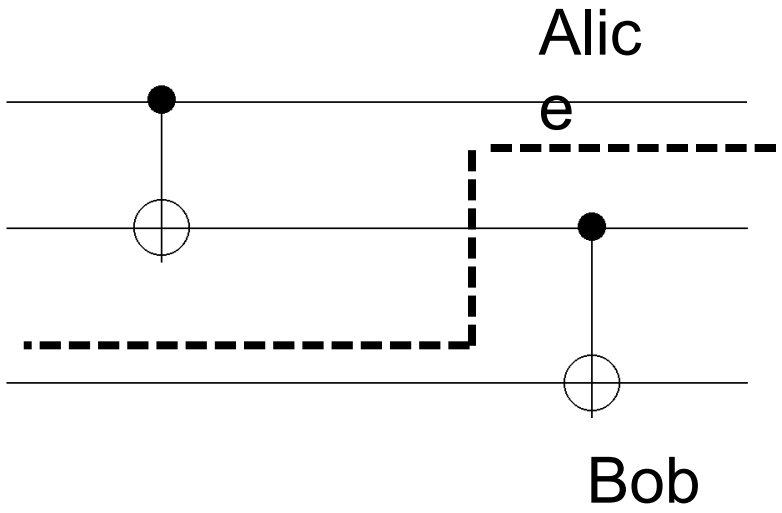
Alice XORs her bit v with the correlated bit and sends the result to Bob.

Bob XORs his correlated bit with the bit Alice sent and thereby obtains a bit with probability vector v .

Classical Teleportation Circuit

$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

$$e = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$



$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

000	$\frac{p}{2}$	$\frac{p}{2}$	$\frac{p}{2}$
001	0	0	0
010	0	0	$\frac{p}{2}$
011	$\frac{p}{2}$	$\frac{p}{2}$	0
100	$\frac{1-p}{2}$	0	0
101	0	$\frac{1-p}{2}$	$\frac{1-p}{2}$
110	0	$\frac{1-p}{2}$	0
111	$\frac{1-p}{2}$	0	$\frac{1-p}{2}$

No information in transmitted bit:

$$\begin{array}{ll} 000 & \frac{p}{2} \\ 001 & 0 \\ 010 & 0 \\ 011 & \frac{p}{2} \\ 100 & 0 \\ 101 & \frac{1-p}{2} \\ 110 & \frac{1-p}{2} \\ 111 & 0 \end{array}$$

transmitted bit

$$\begin{bmatrix} \frac{p}{2}+0+0+\frac{1-p}{2} \\ 0+\frac{p}{2}+\frac{1-p}{2}+0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

And it works:

$$\begin{array}{ll} 000 & \frac{p}{2} \\ 001 & 0 \\ 010 & \frac{p}{2} \\ 011 & 0 \\ 100 & 0 \\ 101 & \frac{1-p}{2} \\ 110 & 0 \\ 111 & \frac{1-p}{2} \end{array}$$

Bob's bit

$$\begin{bmatrix} \frac{p}{2}+\frac{p}{2}+0+0 \\ 0+0+\frac{1-p}{2}+\frac{1-p}{2} \end{bmatrix} = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

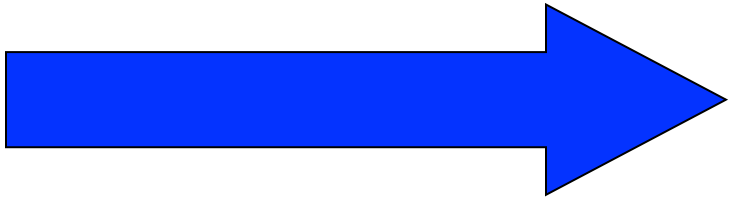
Quantum Teleportation

Alice wants to send her qubit to Bob.
She does not know the wave function of her qubit.

Alic
e



classical communication



Bob



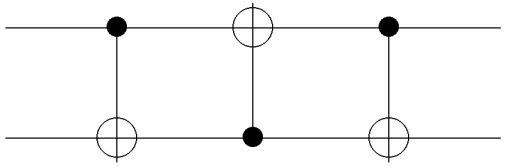
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

allow them to share the entangled state:

$$|e\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi_+\rangle$$

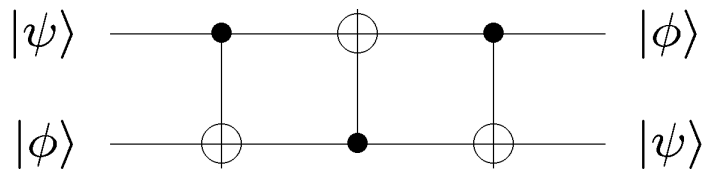
Deriving Quantum Teleportation

Our path: We are going to “derive” teleportation



$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

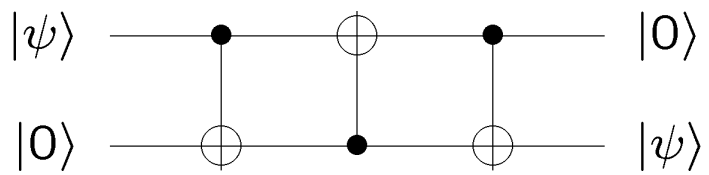
“SWAP”

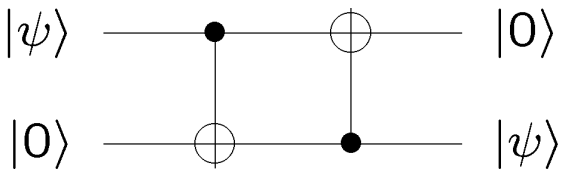


“Alice”

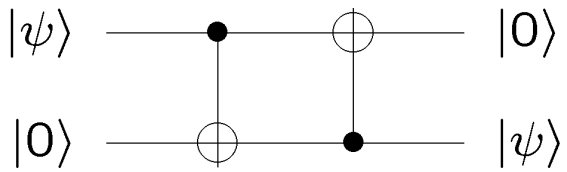
“Bob”

Only concerned with from Alice to Bob transfer

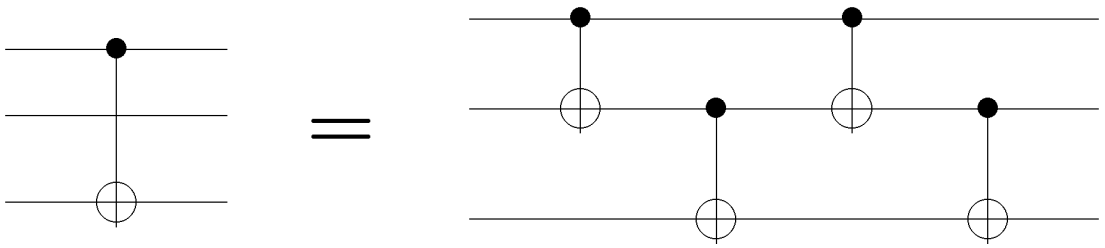




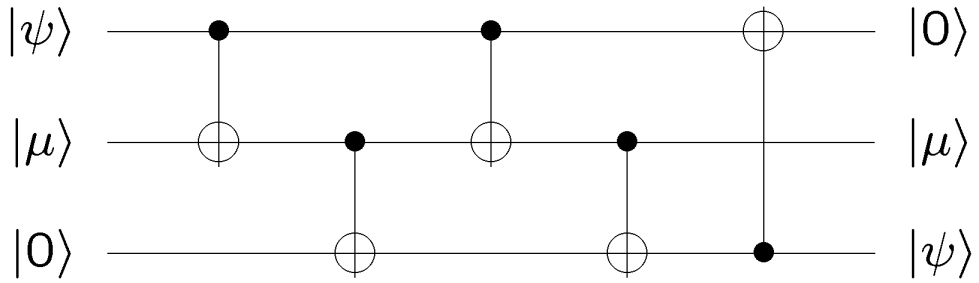
Deriving Quantum Teleportation



Need some way to get entangled states

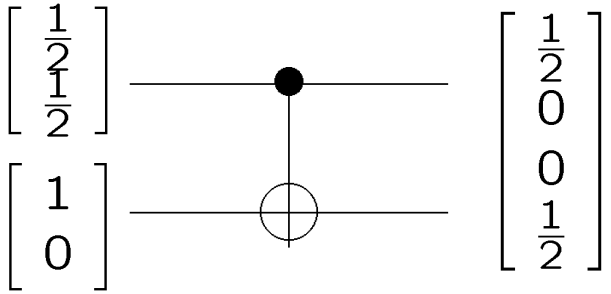


new equivalent circuit:

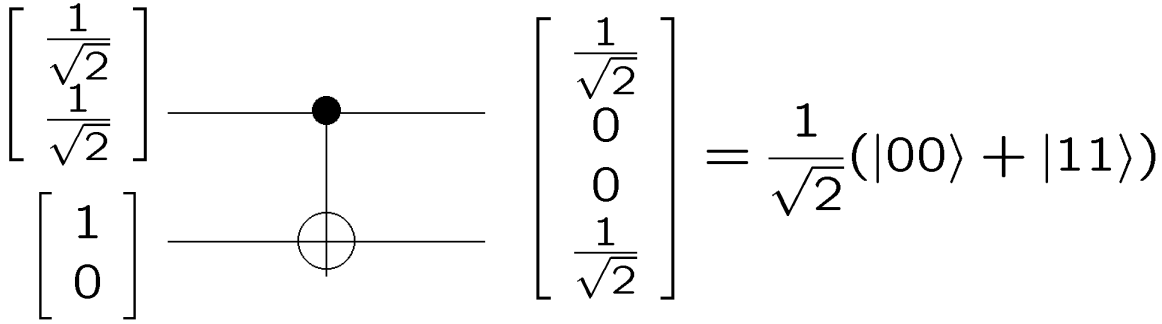


Deriving Quantum Teleportation

How to generate classical correlated bits:



Inspires: how to generate an entangled state:

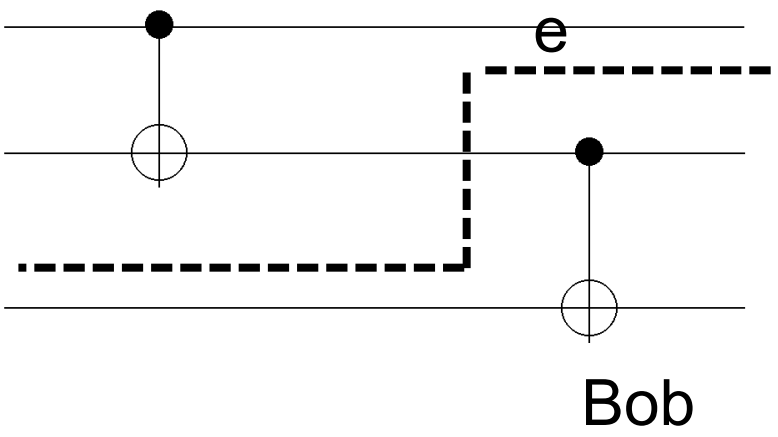


Deriving Quantum Teleportation

Classical Teleportation

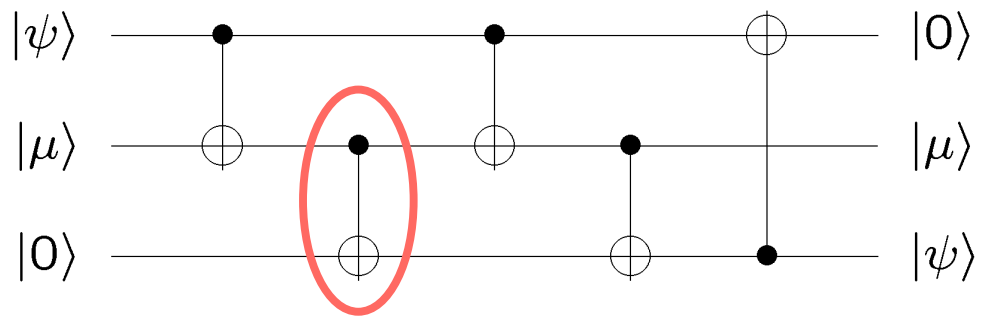
$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

$$e = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$



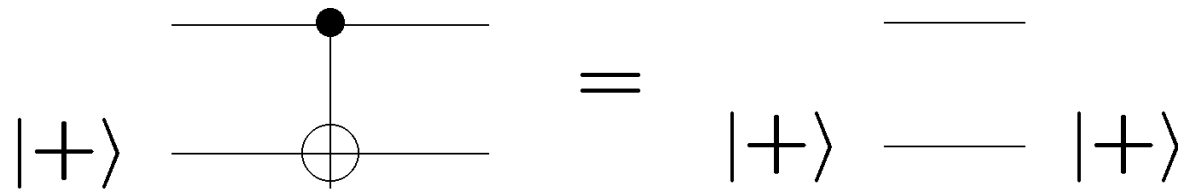
$$r = \begin{bmatrix} \frac{p}{2} \\ \frac{p}{2} \\ \frac{1-p}{2} \\ \frac{1-p}{2} \end{bmatrix}$$

$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

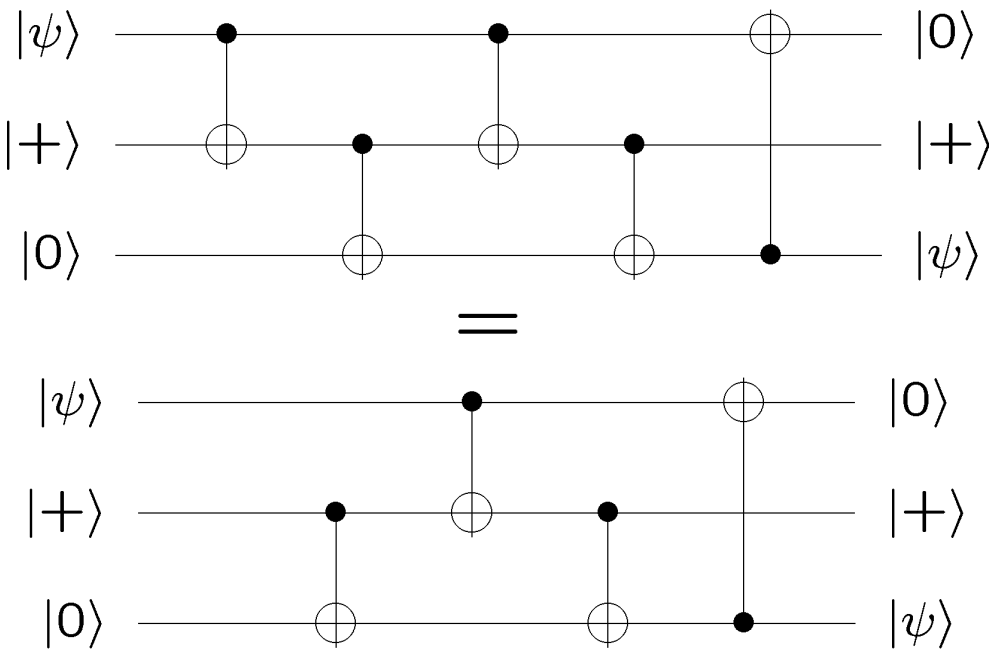


like to use generate entanglement

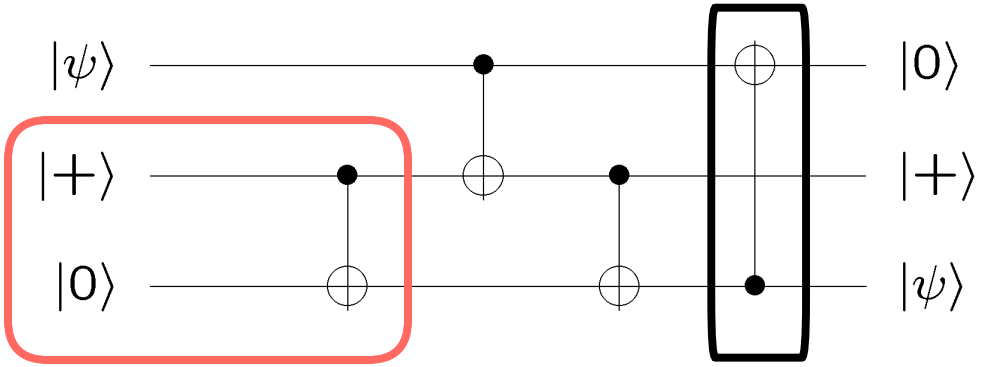
Deriving Quantum Teleportation



$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad X|+\rangle = |+\rangle$$



Deriving Quantum Teleportation



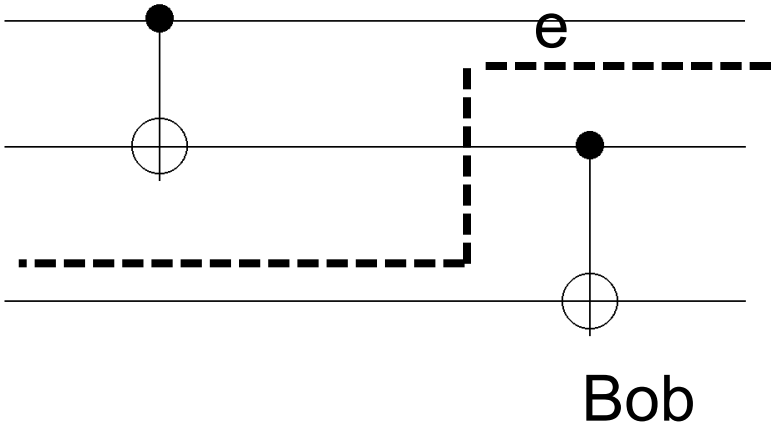
$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

entanglement

?? Acting backwards ??

$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

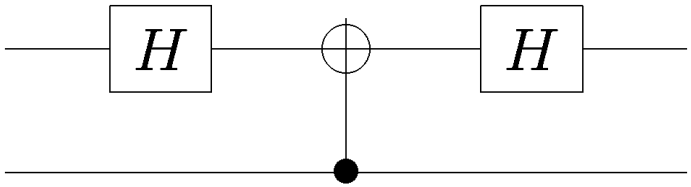
$$e = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$



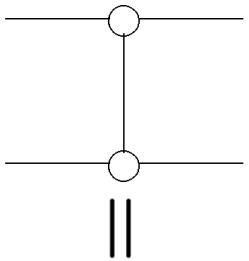
$$r = \begin{bmatrix} \frac{p}{2} \\ \frac{p}{2} \\ \frac{1-p}{2} \\ \frac{1-p}{2} \end{bmatrix}$$

$$v = \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

Deriving Quantum Teleportation



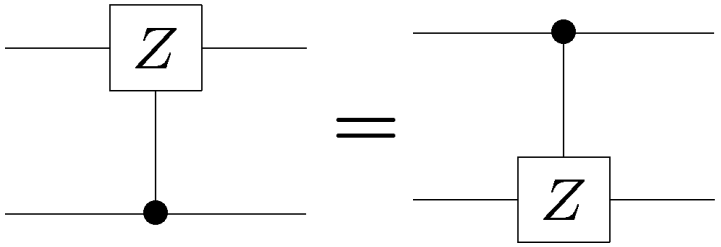
$=$



$=$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$\|$

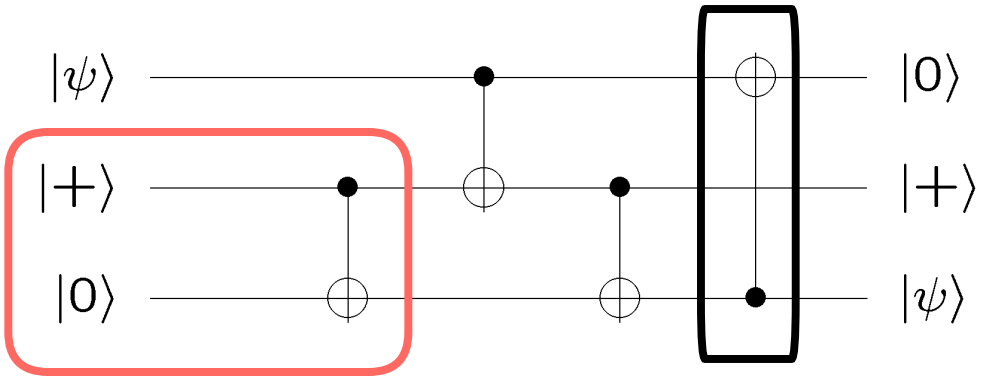


$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$HXH = Z$$

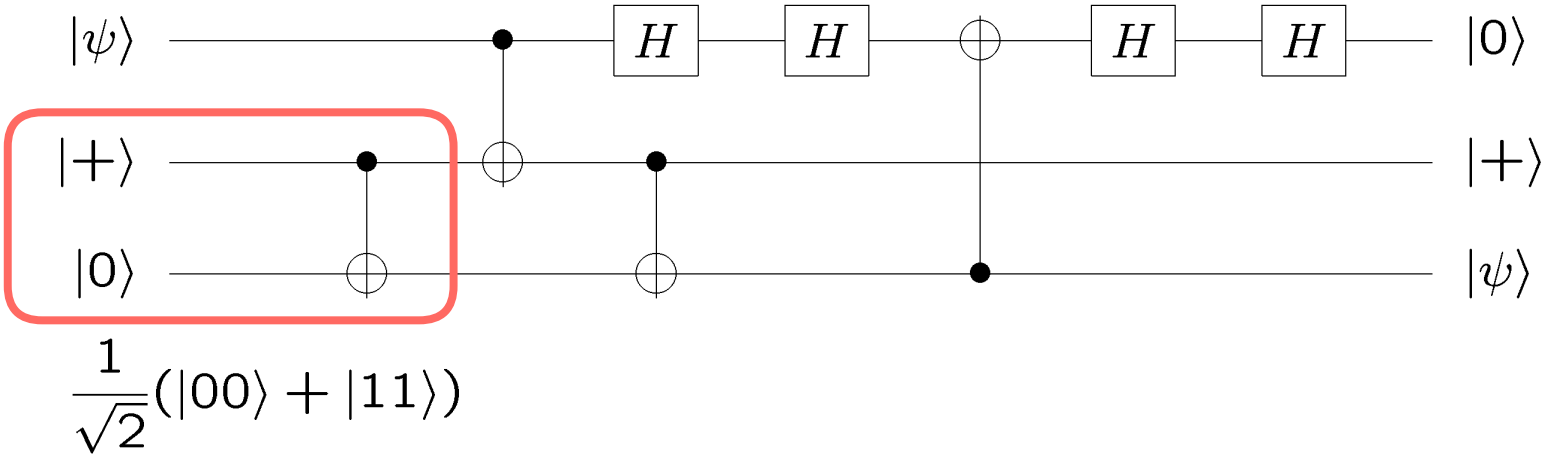
$$H^2 = I$$

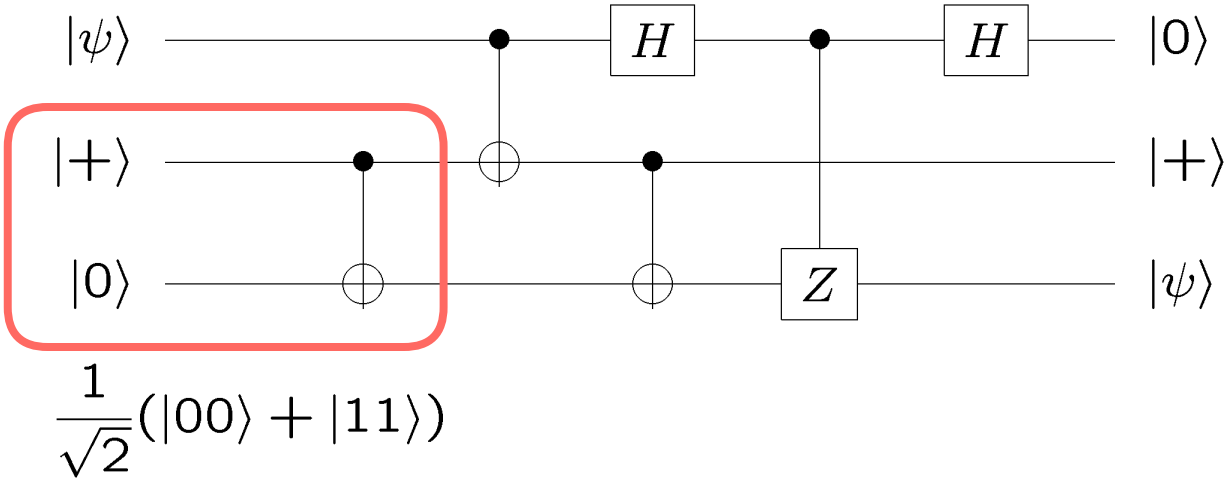
Use to turn around:



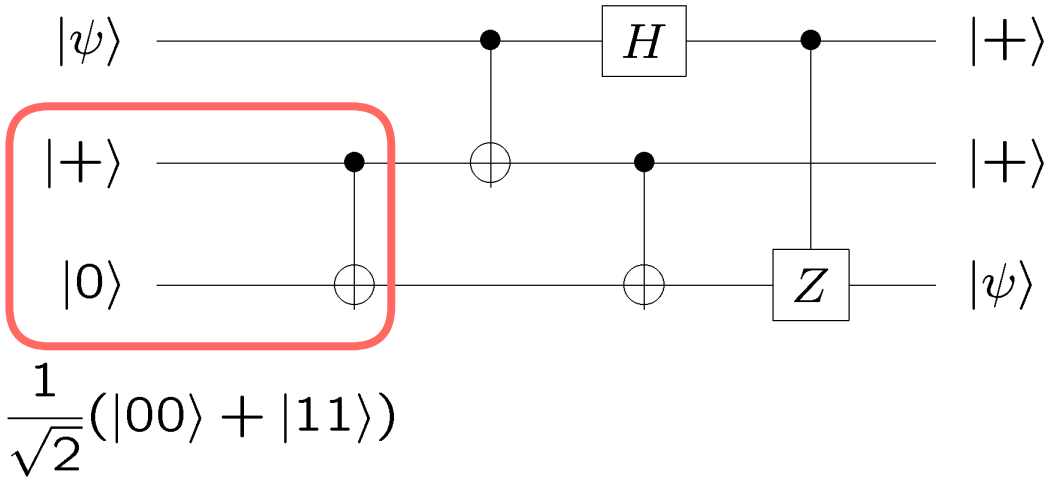
$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

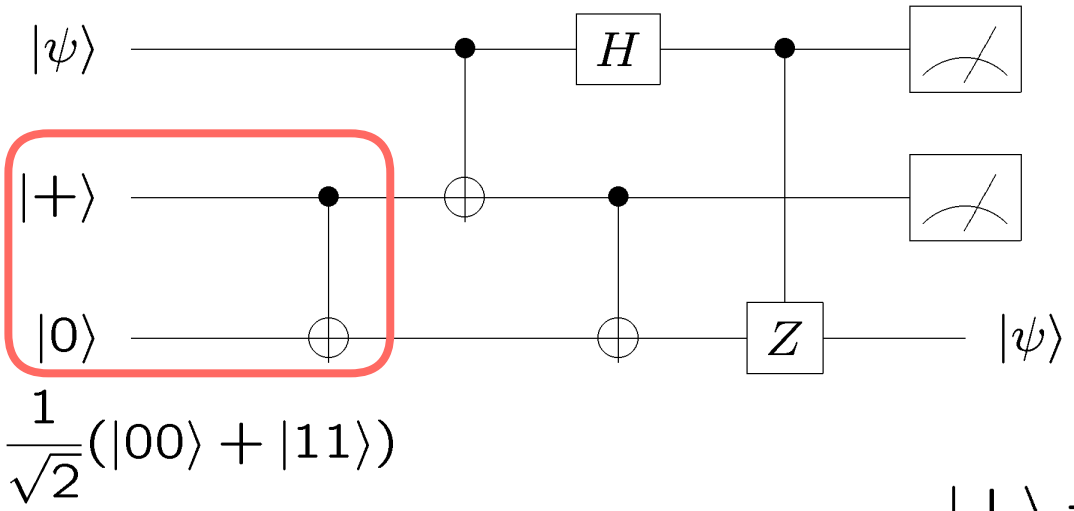
Deriving Quantum Teleportation





Deriving Quantum Teleportation





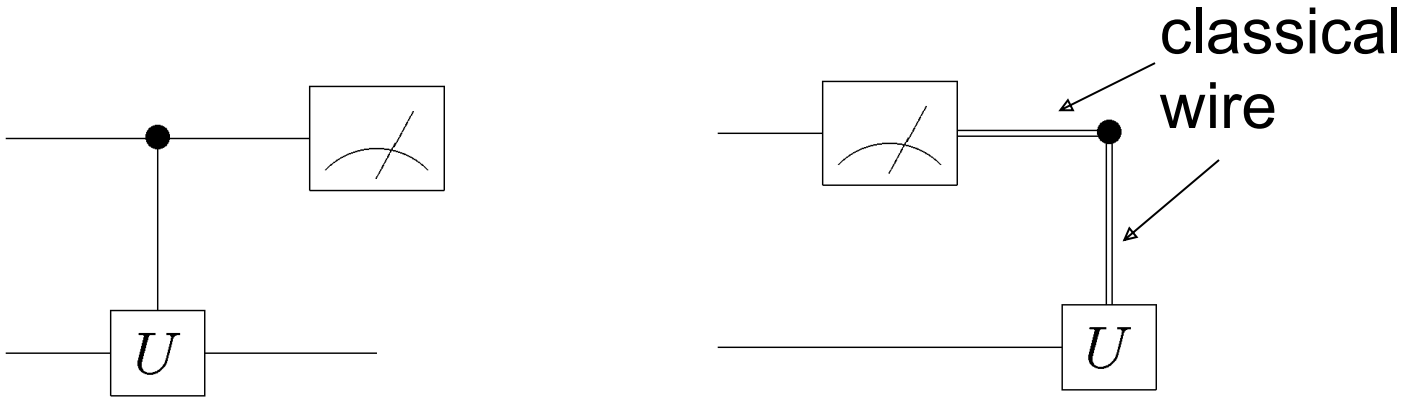
50 % 0, 50 % 1

50 % 0, 50 % 1

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Measurements Through Control

Measurement in the computational basis commutes with a control on a controlled unitary.



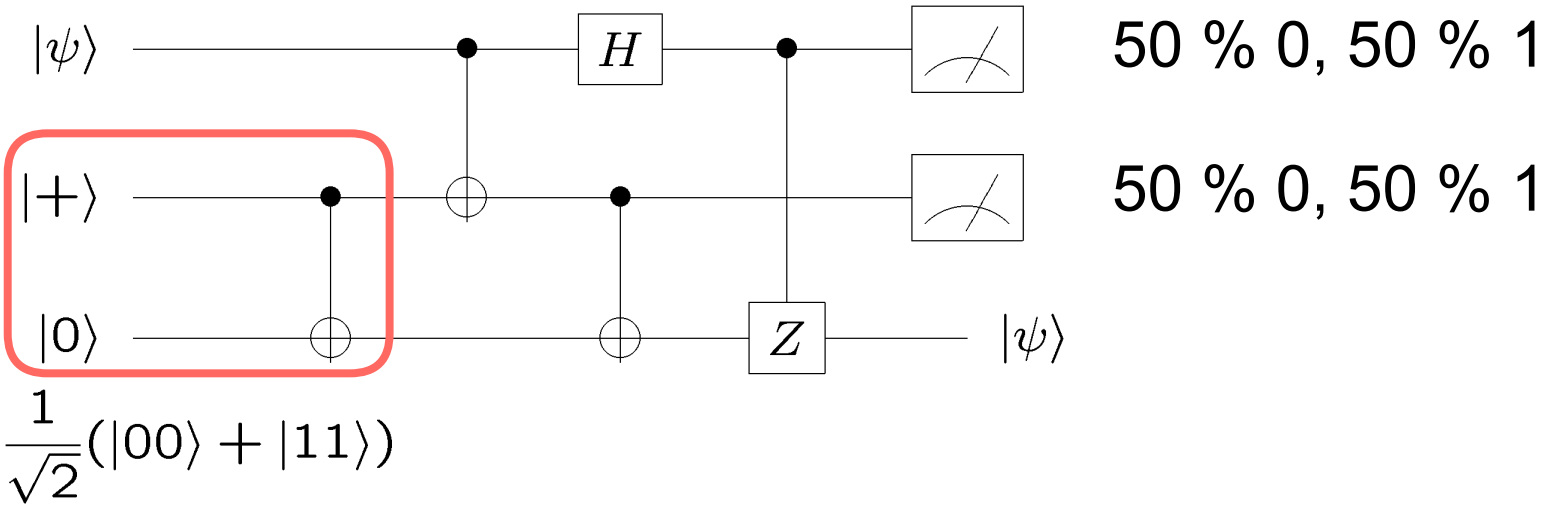
$$(P_0 \otimes I)C_U = C_U(P_0 \otimes I)$$

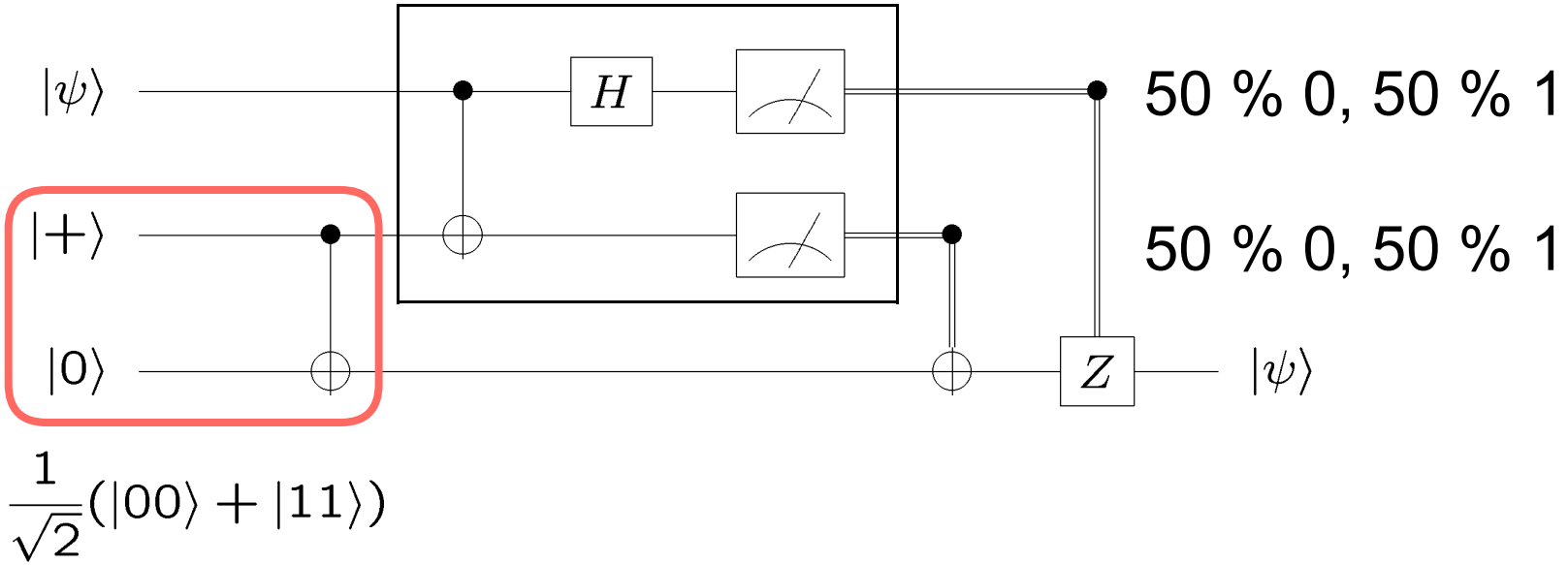
$$(|0\rangle\langle 0| \otimes I)(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U) = (|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U)(|0\rangle\langle 0| \otimes I)$$

$$(P_1 \otimes I)C_U = C_U(P_1 \otimes I)$$

$$(|1\rangle\langle 1| \otimes I)(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U) = (|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U)(|1\rangle\langle 1| \otimes I)$$

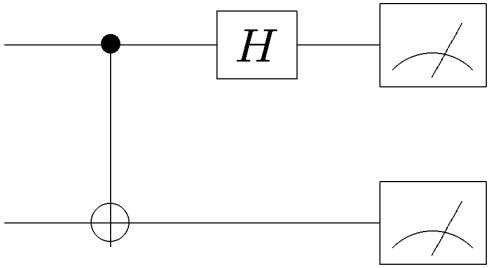
Deriving Quantum Teleportation



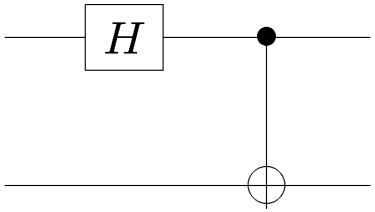


Bell Basis Measurement

Unitary followed by measurement in the computational basis is a measurement in a different basis.



Run circuit backward to find basis:



$$|00\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

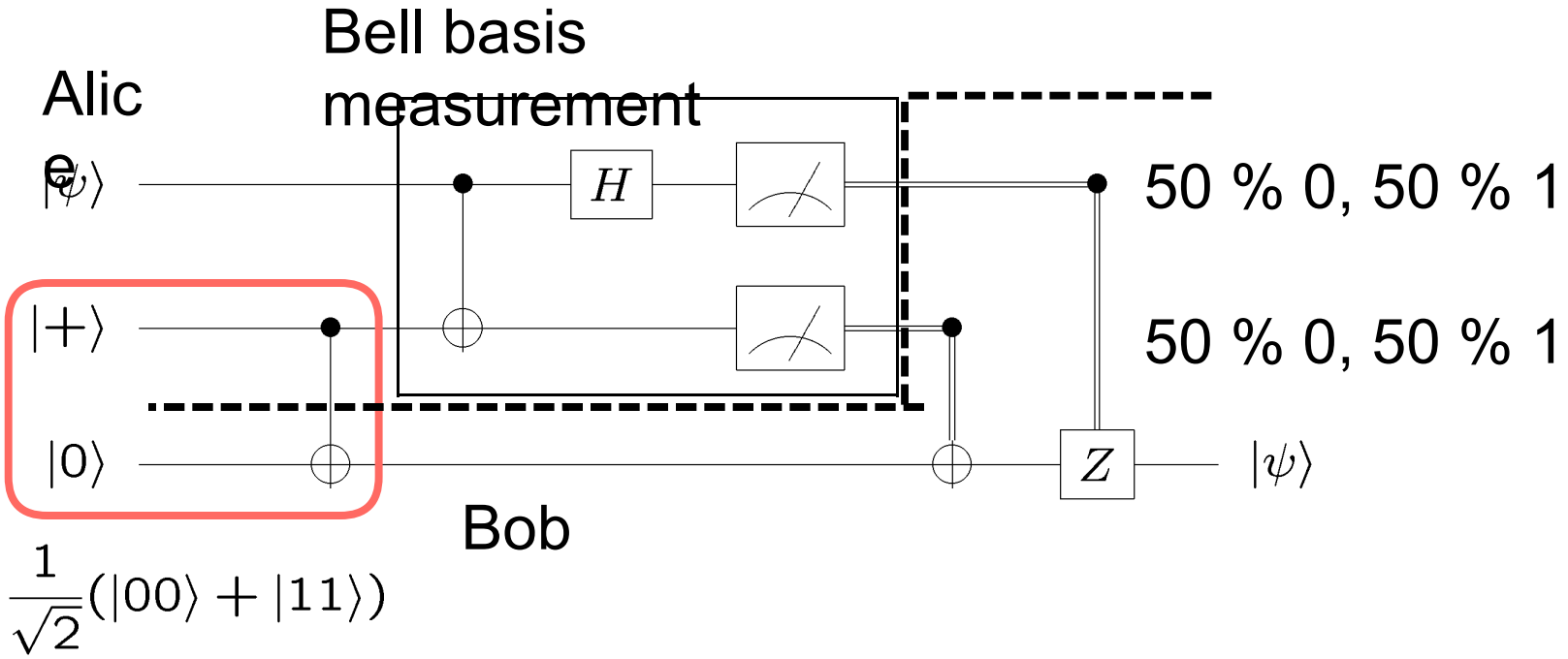
$$|01\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|10\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|11\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Thus we are measuring in the Bell basis.

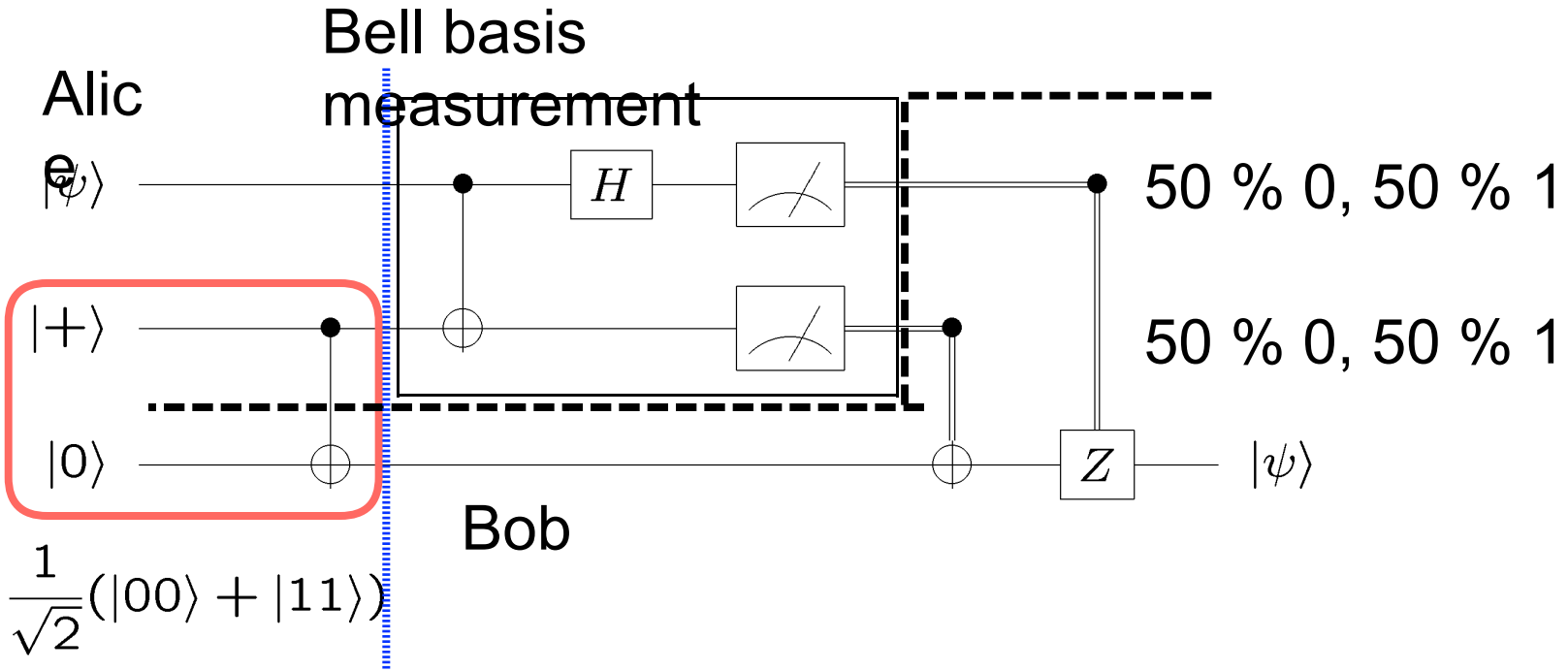
Teleportation



1. Initially Alice has $|\psi\rangle$ and they each have one of the two qubits of the entangled wave function $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
2. Alice measures $|\psi\rangle$ and her half of the entangled state in the Bell Basis.
3. Alice send the two bits of her outcome to Bob who then performs the appropriate X and Z operations to his qubit

In Class Problem 3

Teleportation



$$(\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Teleportation

Bell basis

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
$$|\Phi_-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Computational basis

$$|00\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle + |\Phi_-\rangle)$$
$$|01\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle + |\Psi_-\rangle)$$
$$|10\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$
$$|01\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$

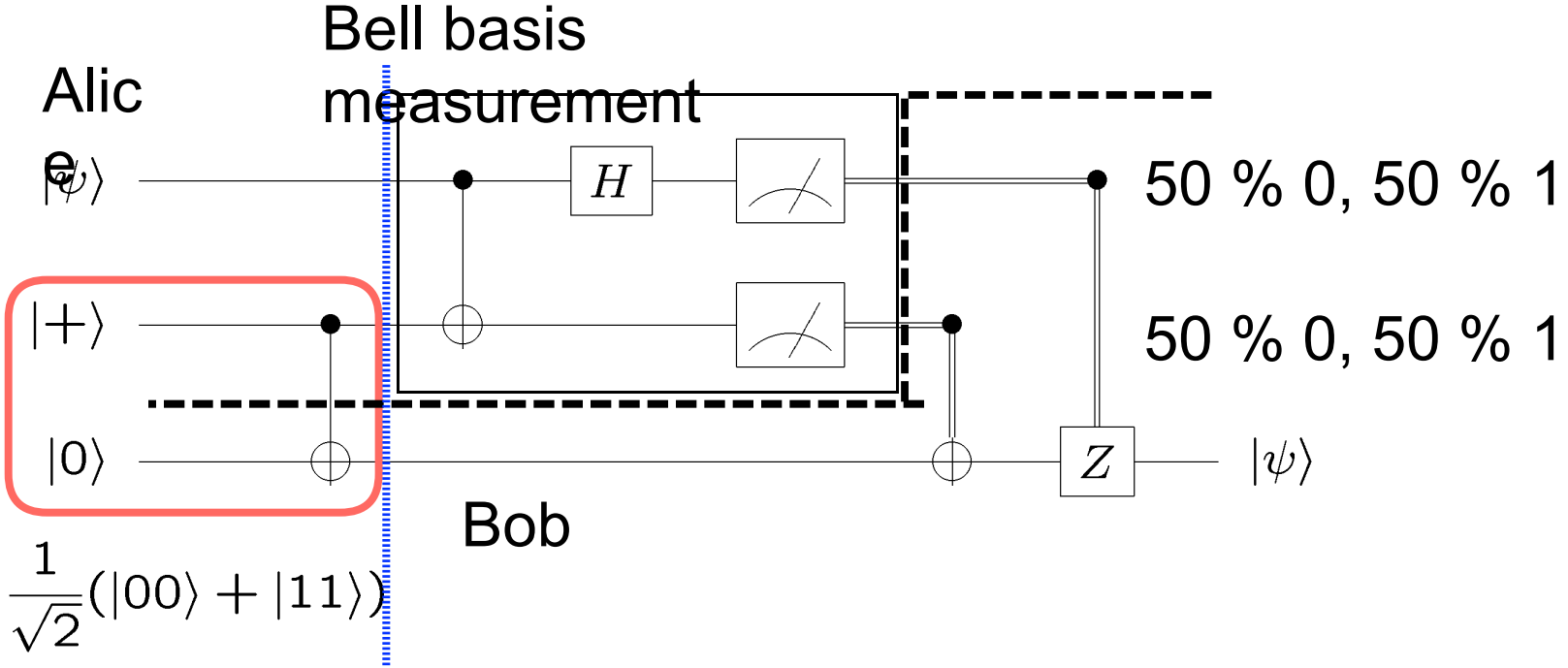
$$(\alpha|0\rangle+\beta|1\rangle)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(\alpha|000\rangle+\alpha|011\rangle+\beta|100\rangle+\beta|111\rangle$$

$$\begin{aligned} &= \frac{1}{2}(\alpha(|\Phi_+\rangle + |\Phi_-\rangle)0\rangle + \alpha(|\Psi_+\rangle + |\Psi_-\rangle)|1\rangle \\ &\quad +\beta(|\Psi_+\rangle - |\Psi_-\rangle)|0\rangle + \beta(|\Phi_+\rangle - |\Phi_+\rangle)|1\rangle \\ &= \frac{1}{2}\Big[|\Phi_+\rangle(\alpha|0\rangle) + \beta|1\rangle + |\Phi_-\rangle(\alpha|0\rangle - \beta|1\rangle) \\ &\quad +|\Psi_+\rangle(\alpha|1\rangle + \beta|0\rangle) + |\Psi_-\rangle(\alpha|1\rangle - \beta|0\rangle)\Big] \end{aligned}$$

Teleportation

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \rightarrow |00\rangle \quad |\Phi_-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \rightarrow |10\rangle$$

$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \rightarrow |01\rangle \quad |\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \rightarrow |11\rangle$$



$$(\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{2}\Big[|\Phi_+\rangle(\alpha|0\rangle) + \beta|1\rangle + |\Phi_-\rangle(\alpha|0\rangle - \beta|1\rangle)$$
$$+|\Psi_+\rangle(\alpha|1\rangle + \beta|0\rangle) + |\Psi_-\rangle(\alpha|1\rangle - \beta|0\rangle)\Big]$$

Teleportation



Alice



Bob

$\cong$

$|\psi\rangle$

Pauli

2 bits

Bell measurement

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$



$|\psi\rangle$

Alice



Bob

Teleportation

1 qubit = 1 ebit + 2 bits

Teleportation says we can replace transmitting a qubit with a shared entangled pair of qubits plus two bits of classical communication.

Superdense Coding

Next we will see that

2 bits = 1 qubit + 1 ebit

Bell Basis

The four Bell states can be turned into each other using operations on only one of the qubits:

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$(X\otimes I)|\Phi_+\rangle = (X\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(|10\rangle+|01\rangle) = |\Psi_+\rangle$$

$$(Z\otimes I)|\Phi_+\rangle = (Z\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(|00\rangle-|11\rangle) = |\Phi_-\rangle$$

$$(ZX\otimes I)|\Phi_+\rangle = (ZX\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(-|10\rangle+|01\rangle) = |\Psi_-\rangle$$

Superdense Coding

Suppose Alice and Bob each have one qubit and the joint two qubit wave function is the entangled state

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice wants to send two bits to Bob. Call these bits b_1 and b_2

Alice applies the following operator to her qubit $Z^{b_2} X^{b_1}$

Alice then sends her qubit to Bob.

Bob then measures in the Bell basis to determine the two bits

2 bits = 1 qubit + 1 ebit

Superdense Coding

Initially $|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

:

Alice applies the following operator to her qubit: $Z^{b_2}X^{b_1}$

$$(Z^{b_2}X^{b_1} \otimes I)|\Phi_+\rangle$$

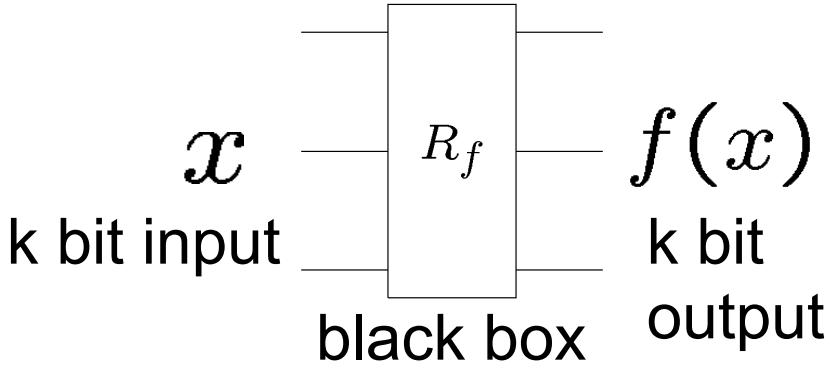
$$\begin{aligned}
b_1 = 0, b_2 = 0 &\qquad |\Phi_+\rangle \\
b_1 = 0, b_2 = 1 &\qquad (Z \otimes I)|\Phi_+\rangle = |\Phi_-\rangle \\
b_1 = 1, b_2 = 0 &\qquad (X \otimes I)|\Phi_+\rangle = |\Psi_+\rangle \\
b_1 = 1, b_2 = 1 &\qquad (ZX \otimes I)|\Phi_+\rangle = |\Psi_-\rangle
\end{aligned}$$

Bob can uniquely determine which of the four states he has and thus figure out Alice's two bits!

Quantum Algorithms

Classical Promise Problem Query Complexity

Given: A black box which computes some function



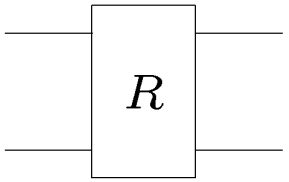
Promise: the function belongs to a set $\mathcal{S}$ which is a subset of all possible functions.

Properties: the set $\mathcal{S}$ can be divided into disjoint subsets $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m$

Problem: What is the minimal number of times we have to use (query) the black box in order to determine which subset $\mathcal{S}_i$ the function belongs to?

Example

Suppose you are given a black box which computes one of the following four reversible classical gates:



2 bits input

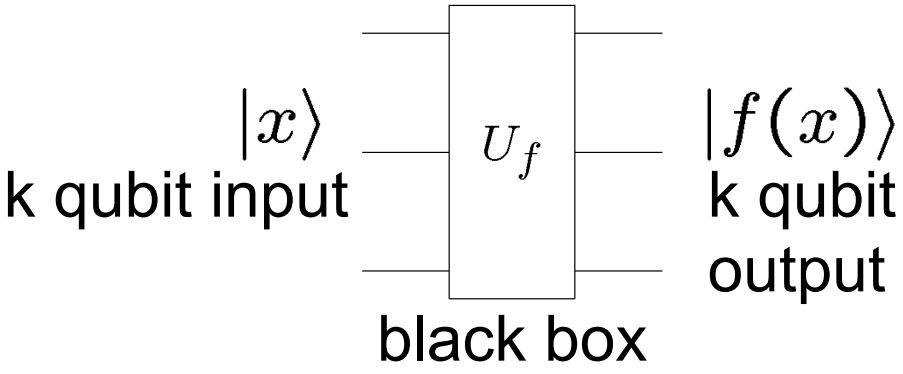
2 bits output

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
	$\mathcal{S}_1$		$\mathcal{S}_2$

Deutsch's (Classical) Problem: What is the minimal number of times we have to use this black box to determine whether we are given one of the first two or the second two functions?

Quantum Promise Query Complexity

Given: A quantum gate which, when used as a classical device computes a reversible function



Promise: the function belongs to a set $\mathcal{S}$ which is a subset of all possible functions.

Properties: the set $\mathcal{S}$ can be divided into disjoint subsets $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m$

Problem: What is the minimal number of times we have to use (query) the quantum gate in order to determine which subset $\mathcal{S}_i$ the function belongs to?